Ocean Monjur

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Research Interests

LLM efficiency, Agentic systems, Hyperspectral Imaging

Education

- **University of South Florida** — Tampa, FL, United States
  PhD. in Computer Science and Engineering Engineering — *Jan 2026 - Current*
  Advisor: Dr. Anshuman Chhabra

- **University of Illinois Urbana-Champaign** — Champaign, IL, United States
  M.Sc. in Agricultural and Biological Engineering;CGPA: 3.81 / 4.00 — *Jan 2024 - Dec 2025*
  Advisor: Dr. Mohammed Kamruzzaman
  Thesis: *Deep learning applications in hyperspectral imaging for agriculture: image reconstruction and model design for quality prediction*

- **Islamic University of Technology** — Gazipur, Bangladesh
  B.Sc. in Computer Science and Engineering; CGPA: 3.57 / 4.00 — *Jan 2019 - May 2023*
  Thesis: *Modifying the internal structure of object detector necks to improve detection and segmentation performance*

Research Experience

- **University of South Florida** — Tampa, FL, United States
  Graduate Research and Teaching Assistant — *Jan 2026 - Present*
  - **LLM efficiency**: Working with pruning methods for LLMs to reduce parameters while keeping similar performance in Test-Time Scaling tasks.
  - **Agentic Systems**: Identifying vulnerabilities in multi-agent systems as well as improving robustness across different agentic tasks.

- **University of Illinois Urbana-Champaign** — Champaign, IL, United States
  Graduate Research Assistant — *Jan 2024 - Dec 2025*
  - **Hyperspectral Image Reconstruction**: Experimenting with different Hyperspectral Image Reconstruction models to convert RGB images into Hyperspectral Images for Agricultural Applications.
  - **Moisture Prediction using Optical Sensing and Machine Learning**: Using optical sensing i.e. Hyperspectral images and Machine Learning to detect current moisture level across different biological samples

- **University of Illinois Urbana-Champaign**
  Remote Unpaid Research Collaboration — *May 2023 - November 2023*
  - **Literature Review**: Performed a comprehensive literature review of Hyperspectral Image reconstruction models and selected methods suited for agricultural applications

- **Computer Vision Lab, Islamic University of Technology** — Gazipur, Bangladesh
  Student Researcher — *Nov 2021 - May 2023*
  - **Complementarity of ConvNets, Vision Transformers and MLP mixers**: Exploratory analysis of a large combinations of ConVNets,ViTs and MLP mixers to find complementarity characteristics which might increase performance when combined together. This project is supervised by Dr. Niamul Quader
  - **Object Detection and Segmentation**: Experimented with Object Detectors Necks to increase better feature fusion with lower-level high-resolution feature maps and higher-level semantic feature maps to improve performance on small objects.

- **Apurba Technologies Ltd** — Mohakhali, Dhaka
  Part-Time Machine Learning Engineer — *Mar 2023 - May 2023*
  - **Maintaining and Updating the OCR pipeline:**: My primary job is to keep the Core OCR pipeline operational. Commit changes to the pipeline when required and maintain the servers in which the OCR models operate.

- **ACI Motors** — Dhaka, Bangladesh
  Intern — *Oct 2021 - Dec 2021*
  - **Data analysis:**: My job was to try and make out actionable insights from farmer data.

Publications

1. **O. Monjur**, S. K. Nahin, and A. Chhabra. Revisiting the effectiveness of llm pruning for test-time scaling. *EMNLP (Findings)*, 2026
2. **O.Monjur**, M. T. Ahmed, M. W. Ahmed, and M. Kamruzzaman. Agro-net: A convolution-attention fusion based hyperspectral model for agro-food quality assessment. In *Proceedings of the Computer Vision and Pattern Recognition Conference Workshops*, pages 474–481, 2025
3. **O. Monjur**, M. T. Ahmed, G. Chowdhary, and M. Kamruzzaman. Agro-hsr: The first large-scale agricultural-focused hyperspectral dataset for deep learning-based image reconstruction and quality prediction. *Computers and Electronics in Agriculture*, 239:111103, 2025
4. T. D. Belay, S. K. Nahin, I. A. Azime, **Monjur, Ocean**, M. Rei, C. Biemann, S. H. Muhammad, S. M. Yimam, and A. Chhabra. Afrilangtutor: Advancing language tutoring and culture education in low-resource languages with large language models. *EMNLP (Findings)*, 2026
5. M. T. Ahmed, **O. Monjur**, A. Khaliduzzaman, and M. Kamruzzaman. A comprehensive review of deep learning-based hyperspectral image reconstruction for agri-food quality appraisal. *Artificial Intelligence Review*, 58(4):96, 2025
6. M. T. Ahmed, **O. Monjur**, and M. Kamruzzaman. Deep learning-based hyperspectral image reconstruction for quality assessment of agro-product. *Journal of Food Engineering*, 382:112223, 2024
7. M. T. Ahmed, M. W. Ahmed, **O. Monjur**, J. L. Emmert, G. Chowdhary, and M. Kamruzzaman. Hyperspectral image reconstruction for predicting chick embryo mortality towards advancing egg and hatchery industry. *Smart Agricultural Technology*, 9:100533, 2024
8. **O. Monjur**, R. B. Preo, A. B. Shams, M. Raihan, M. Sarker, and F. Fairoz. Covid-19 prognosis and mortality risk predictions from symptoms: A cloud-based smartphone application. *BioMed*, 1(2):114–125, 2021

Publications - Under Review/Preprints

1. O. Monjur, D. A. Nisar, S. K. Nahin, T. D. Belay, S. Mishra, I. A. Azime, and A. Chhabra. Lowagent: A low-resource multilingual benchmark for web agents. *arXiv preprint arXiv:2610.27205*, 2026 **Submitted to a double blinded conference.**
2. M. Bashar, **O. Monjur**, S. Islam, M. G. Shams, and N. Quader. Exploring synergistic ensemble learning: Uniting cnns, mlp-mixers, and vision transformers to enhance image classification. *arXiv preprint arXiv:2504.09076*, 2025 **Submitted to a double blinded conference.**
3. A. B. Shams, M. Raihan, M. Sarker, M. Khan, M. Uddin, **O. Monjur**, and R. B. Preo. Telehealthcare and telepathology in pandemic: A noninvasive, low-cost micro-invasive and multimodal real-time online application for early diagnosis of covid-19 infection. *arXiv preprint arXiv:2109.07846*, 2021

Reviewing Experience

1. INTERNATIONAL JOURNAL OF DIGITAL EARTH (2026): 1 paper
2. CVPR Workshops V4A 2024, 2025: 5 papers

Honors and Awards

1. CARD-IAB Meeting May 2024 Poster Presentation Award: 3rd Place
2. CARD-IAB Meeting October 2024 Poster Presentation Award: 2nd Place
3. CARD-IAB Meeting May 2025 Poster Presentation Award: Honorable Mention

Projects

- **Image Classification of Stroke Blood Clot Origin**: Developed an Ensemble of Classifiers of CoAtNet and ResNet-101 for the Kaggle competition. The dataset for this competition comprises over a thousand high-resolution whole-slide digital pathology images. Each slide depicts a blood clot from a patient that had experienced an acute ischemic stroke. Our task is to classify the etiology (CE or LAA) of the slides in the test set for each patient.
- **Automatic Speech Recognition**: Worked on fine tuning automatic speech recognition models to improve Bengali speech recognition for specific local dialects. Finished in the top-10 out of 54 competing teams.

Skills Summary

- **Programming Languages**: Python, C++, Matlab
- **Tools**: Git, Docker
- **Frameworks**: Pytorch, Tensorflow